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A Theory of Cash Flow-Based Financing with Distress Resolution

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1 Big picture

- **Question.** What determines whether firms borrow against collateral value alone (asset-based financing) or against going-concern cash flows (cash flow-based financing), and how are those financing constraints tied to bankruptcy and distress resolution?
- **Setting.** The paper is a dynamic contracting theory. A firm with risky cash flows, costly internal cash, and imperfect access to outside equity signs a long-term contract with an intermediary that can always provide liquidity but has limited risk-bearing capacity.
- **Core challenge.** Standard financing models often take debt structure or bankruptcy as given. This paper instead wants those objects to emerge endogenously from one problem: how should a financially constrained firm optimally share liquidity risk with a lender over time?
- **Main conceptual contribution.** The paper unifies *financing constraints*, *liquidity management*, and *bankruptcy*. The financing constraint is not just a collateral number imposed from outside; it is the value that the intermediary can rationally promise itself under the optimal contract.
- **Main mechanism.** Future refinancing opportunities matter today. If the intermediary expects the firm to survive distress and later sell itself to new equity investors, then the intermediary values the firm as a going concern and is willing to extend more secured financing today. That is what makes financing cash-flow-based rather than purely asset-based.
- **State variable.** The paper shows that the relevant state is **net liquidity**,

$$C_t \equiv M_t - Y_t,$$

cash held inside the firm minus promised continuation value to the intermediary. This single state variable summarizes both cash reserves and debt obligations.

- **Main theoretical result.** The optimal contract is implemented by **equity + senior-secured debt + an unsecured credit line**. Senior-secured debt finances promised repayments and is always repaid in full; the unsecured credit line absorbs risk and is allowed to default in distress.
- **Key financing result.** Financing capacity can be:
 - **asset-based:** pinned down by liquidation value L ;
 - **cash-flow-based:** pinned down by the intermediary's endogenous valuation of the firm as a going concern.

Which regime obtains depends on firm profitability and volatility, intermediary risk-bearing capacity, and access to outside equity.

- **Key bankruptcy result.** Bankruptcy is not just liquidation. It is an optimal *risk-sharing device*. Chapter 11 corresponds to continuation with dilution of equity and partial default on unsecured debt; Chapter 7 corresponds to liquidation after the firm's financing capacity is exhausted.

- **Main complementarity result.** Unsecured and secured debt are complements. Because unsecured debt absorbs risk in bad states, it protects secured debt and can raise secured debt capacity above liquidation value.
- **Second complementarity result.** Debt and equity are dynamic complements. Better future access to equity financing increases current secured debt capacity because lenders expect to be repaid when refinancing opportunities arrive.
- **Main comparative-statics result.** Cash flow-based financing becomes more likely when:
 - profitability is higher,
 - cash-flow volatility is lower,
 - intermediary risk-bearing capacity is higher,
 - and access to outside equity is better.
- **Economic takeaway.** A firm's ability to borrow against cash flows is inseparable from how distress will be resolved. Chapter-11-style continuation can itself create the collateral value that makes cash-flow-based lending feasible.

2 Model environment and key objects

2.1 Firm, investors, intermediary, and frictions

- Time is continuous and infinite.
- The firm's cash flows evolve as

$$dX_t = \mu dt + \sigma dZ_t, \quad (1)$$

where μ is the cash-flow drift and σ is volatility.

- Shareholders are hand-to-mouth: they can consume payouts but cannot inject funds into the firm.
- Access to new outside equity investors arrives only at Poisson times with intensity π .
- An intermediary can transfer funds to the firm at any time, but bearing risk is costly for the intermediary.

Interpretation. The firm's central problem is a liquidity problem. Cash-flow shortfalls happen continuously, but new equity can only be raised intermittently.

2.2 Cash holdings and the cost of liquidity

The firm can retain internal cash M_t , but cash is costly: it earns only $r - \lambda$ rather than the full discount rate r , where $\lambda > 0$ is the carry cost of cash. Cash evolves as

$$dM_t = dX_t + (r - \lambda)M_t dt - dDiv_t - dI_t + \Delta M_t d\Pi_t. \quad (2)$$

Here:

- $dDiv_t$ is payout to shareholders,
- dI_t is the net transfer between the intermediary and the firm,
- ΔM_t is cash raised at refinancing times $d\Pi_t = 1$.

Interpretation. The firm smooths shocks with two costly instruments: internal cash and intermediary finance.

2.3 The intermediary's continuation value and promise keeping

Let Y_t denote the promised continuation value of the intermediary. The intermediary's total compensation process satisfies the promise-keeping condition

$$dY_t + dI_t = rY_t dt + dK_t + \beta_t \sigma dZ_t + \alpha_t (d\Pi_t - \pi dt). \quad (3)$$

The intermediary's required flow compensation is

$$\frac{dK_t}{dt} = k(\alpha_t, \beta_t) = \sigma^2 \left(\frac{\rho r \beta_t^2}{2} \right) + \pi \left(\alpha_t - \frac{1 - e^{-\rho r \alpha_t}}{\rho r} \right). \quad (4)$$

- β_t is the intermediary's exposure to cash-flow risk.
- α_t is the intermediary's exposure to refinancing or capital-supply risk.
- $1/\rho$ is the intermediary's risk-bearing capacity.

Interpretation. The contract can shift two risks to the intermediary: ordinary cash-flow risk and the risk that refinancing opportunities arrive late.

2.4 Autarky value and the role of liquidation

If the intermediary keeps the firm alive without future outside refinancing, its autarky value is

$$Y^A \equiv \frac{\mu}{r} - \frac{\rho}{2} \sigma^2. \quad (5)$$

If $Y^A < L$, the intermediary would rather liquidate the firm immediately and realize liquidation value L .

Interpretation. Y^A is the lender's stand-alone going-concern value. It is already lower than the first-best value μ/r because the intermediary dislikes bearing risk.

2.5 Equity value and the firm's optimization problem

The shareholders' value is

$$P_t = E_t \left[\int_t^\tau e^{-r(s-t)} (dDiv_s - \Delta M_s d\Pi_s) \right], \quad (6)$$

and the contract chooses payouts, intermediary transfers, and refinancing subject to:

$$dDiv_t \geq 0, \quad \Delta M_t \leq P_t. \quad (7)$$

The second condition is limited liability: the firm cannot issue more equity than the post-refinancing firm is worth.

Interpretation. Equity issuance dilutes existing shareholders, and because shareholders cannot contribute cash, they can promise only what new investors are willing to buy.

2.6 Net liquidity as the sufficient state variable

The crucial simplification is that the contract can be summarized by

$$C_t \equiv M_t - Y_t. \quad (8)$$

The paper interprets C_t as **excess liquidity**: cash on hand minus the intermediary's promised continuation value.

Two immediate constraints follow:

$$Y \geq \max\{0, -C\}. \quad (9)$$

So promised payments cannot be negative, and if net liquidity is negative, the firm must owe at least $-C$ to the intermediary.

Interpretation. High C means the firm is flush with liquidity; low or negative C means the firm is dependent on the intermediary.

2.7 Refinancing accounting and the key constraint

Let C^* be post-refinancing net liquidity. The cash raised at refinancing satisfies

$$\Delta M = C^* - C + \alpha. \quad (10)$$

Combining this accounting identity with limited liability gives the central refinancing constraint:

$$\alpha \leq (P(C^*) - C^*) + C. \quad (11)$$

Interpretation. The firm cannot shift arbitrarily much refinancing risk onto the intermediary. Payments promised to the intermediary at refinancing cannot exceed the resale value of the firm plus available cash.

2.8 Key objects to keep straight

- $M(C)$: internal cash held by the firm.
- $Y(C)$: promised continuation value owed to the intermediary.
- $C = M - Y$: net liquidity.

- $\alpha(C)$: capital-supply risk shifted to the intermediary.
- $\beta(C)$: cash-flow risk shifted to the intermediary.
- $\bar{C}$: upper payout boundary where dividends are paid and refinancing targets reset.
- $\underline{C}$: lower boundary where the firm either liquidates or is continued by the intermediary.
- C^S : lower continuation boundary under nonliquidation.

Interpretation. Almost every economic result in the paper can be read by tracking what happens to these objects as C moves from high liquidity to severe distress.

3 Core model and theoretical results (all equations + interpretation)

3.1 The HJB and the payout boundary

The shareholders' value function solves the HJB

$$rP(C) = \max_{\beta, Y} \left\{ P'(C) \left[\mu + (r - \lambda)C - \lambda Y - \frac{\sigma^2 \rho r}{2} \beta^2 \right] + P''(C) \frac{\sigma^2}{2} (1 - \beta)^2 \right\} \quad (12)$$

$$+ \pi \max_{C^*, \alpha} \left\{ P'(C) \left(\frac{1 - e^{-\rho r \alpha}}{\rho r} \right) + \left[(P(C^*) - C^*) - (P(C) - C) - \alpha \right] \right\}, \quad (13)$$

subject to the deferred-payment and refinancing constraints.

At the upper boundary $\bar{C}$, the firm pays dividends, so smooth pasting and super-contact imply

$$P'(\bar{C}) = 1, \quad P''(\bar{C}) = 0. \quad (14)$$

Interpretation. The shareholders act as if risk-averse because liquidation is inefficient, outside equity arrives only intermittently, and intermediary funds are costly. So they hoard cash until the payout boundary is reached.

3.2 Proposition 1: value function and dividends

Proposition 1 says that equity value can be written as a function of C alone, $P_t = P(C_t)$, and that $P(C)$ is strictly concave on $(\underline{C}, \bar{C})$ with $P'(C) > 1$ for $C < \bar{C}$.

Main implications:

- dividends are zero in the interior;
- the firm reflects at $\bar{C}$ through payout;
- the payout boundary is strictly positive: $\bar{C} > 0$.

Interpretation. Even a liquidity-constrained firm optimally keeps a strictly positive cash cushion before distributing anything.

3.3 Optimal controls: refinancing target, promised value, and risk sharing

The optimal refinancing target is simply the payout boundary:

$$C^*(C) = \bar{C}. \quad (15)$$

So every refinancing resets the firm to the same target net-liquidity position.

Because cash is costly, the contract uses the minimum promised continuation value consistent with feasibility:

$$Y(C) = \max\{-C, 0\}, \quad M(C) = \max\{C, 0\}. \quad (16)$$

Thus the firm either holds cash or owes promised value to the intermediary, but not both at once.

The optimal cash-flow risk share is

$$\beta(C) = \frac{P''(C)}{P''(C) - \rho r P'(C)} \in [0, 1]. \quad (17)$$

When the refinancing constraint does not bind, the unconstrained optimal capital-supply risk share is

$$\alpha^U(C) = \frac{\ln P'(C)}{\rho r}. \quad (18)$$

With limited liability, the actual choice is

$$\alpha(C) = \min \{ \alpha^U(C), P(C) - C + \underline{C} \} \geq 0. \quad (19)$$

Interpretation.

- $\beta(C)$ rises when the firm is more constrained because the shareholders want to offload more cash-flow risk.
- $\alpha(C)$ is a refinancing-risk-sharing device: it shifts payments from constrained states into future unconstrained refinancing states.
- The limited-liability constraint matters most in distress, because that is where the firm would like to promise the intermediary a lot but cannot credibly do so.

3.4 The lower boundary and financing capacity

There are two possibilities for the lower boundary.

Case 1: liquidation. The firm liquidates at the lowest point consistent with promise keeping:

$$\underline{C} = -L. \quad (20)$$

Case 2: continuation. The intermediary keeps the firm alive at an endogenous continuation boundary C^S , where volatility vanishes because $\beta(C) \rightarrow 1$ and the intermediary absorbs all cash-flow risk.

Lemma 3 gives

$$\underline{C} = \min\{C^S, -L\}, \quad P(\underline{C}) = 0. \quad (21)$$

An approximation for the continuation value at the lower boundary is

$$Y(C^S) = -C^S \approx \frac{r}{r + \pi} Y^A + \frac{\pi}{r + \pi} (P(\bar{C}) - \bar{C}). \quad (22)$$

Interpretation. Cash-flow-based financing is possible because the intermediary values not just current operations but also the future resale value of the firm when equity market access arrives.

3.5 Proposition 2 and Corollary 1: asset-based versus cash-flow-based financing

The firm's financing capacity is

$$\bar{Y} \equiv Y(\underline{C}) = -\underline{C} = \max\{L, -C^S\}. \quad (23)$$

So:

- if $\underline{C} = -L$, financing capacity is **asset-based**;
- if $\underline{C} = C^S$, financing capacity is **cash-flow-based**.

Corollary 1 gives the comparative statics:

- if intermediary risk-bearing capacity is very low (large ρ), financing is asset-based;
- if access to outside equity is very poor (small π), financing is more likely to be asset-based;
- if intermediary risk-bearing capacity is high or refinancing access is good, financing is more likely cash-flow-based.

Interpretation. The lender can extend cash-flow-based finance only when it can bear enough risk and expects the firm to reach future refinancing states often enough.

3.6 Proposition 3: defaultable debt as the state variable implementation

To move from the forward-looking state C to a security that looks like actual debt, the paper defines the cumulative net transfer since last refinancing,

$$T_t := - \int_{\tau^\Pi(t)}^t e^{\int_s^t \hat{r}_u du} dI_s. \quad (24)$$

If $T_t = T(C_t)$ is Markov, then Proposition 3 shows

$$T(C) = \alpha^U(C) + Y(C), \quad (25)$$

and the effective interest rate satisfies

$$\hat{r}(C)T(C) = \frac{\lambda}{\rho r} \mathbf{1}_{\{C \geq 0\}} + rY(C). \quad (26)$$

Interpretation. The optimal contract can be summarized by a face value of debt. That debt may be defaulted on in distress, so defaultable debt emerges directly from the contracting problem.

3.7 From debt overhang to control rights

Define the change in existing shareholders' value upon refinancing by

$$J(C) \equiv (P(\bar{C}) - \bar{C}) - (P(C) - C) - \alpha \geq -P(C). \quad (27)$$

When $J(C) < 0$, existing shareholders lose from refinancing because dilution is severe. This is a financial-overhang or debt-overhang region. The intermediary must therefore receive control rights in those states.

Interpretation. Bankruptcy is needed because equity holders would otherwise refuse to refinance even when continuation is efficient.

3.8 Corollary 2: implementation via secured and unsecured debt

The optimal contract can be implemented with two instruments.

Senior-secured debt.

- Balance: $Y(C)$.
- Always repaid in full.
- Grows at the risk-free rate r .
- Used only when the firm has run out of cash.

Unsecured risky credit line.

- Balance: $\alpha^U(C)$.
- Implements risk-sharing through $\alpha(C)$ and $\beta(C)$.
- Can be partially defaulted on at refinancing and fully defaulted on at liquidation.

The paper writes the dynamics as

$$dY(C) = rY(C)dt - dI^Y(C), \quad (28)$$

$$d\alpha^U(C) = \frac{\lambda}{\rho r} \mathbf{1}_{\{C \geq 0\}} dt - dI^{\alpha^U}(C) - [\alpha^U(C) - \alpha(C)]d\Pi. \quad (29)$$

Interpretation. Secured debt is the safe piece that finances promised repayment capacity; unsecured debt is the risky piece that absorbs bad-state losses and implements state-contingent insurance.

3.9 Why secured and unsecured debt are complements

This is one of the paper's central insights.

- Unsecured debt absorbs losses in distress and buffers negative cash-flow shocks.
- That protection makes secured debt safer.
- Because secured debt becomes safer, the intermediary is willing to lend more against the firm's going-concern value.

Interpretation. Unsecured debt is not merely junior debt sitting below secured debt in priority. It actively creates the conditions under which secured debt can be extended on a cash-flow basis.

3.10 Why debt and equity are dynamic complements

A higher π means more frequent access to new equity investors. That does two things:

- equity statically substitutes for debt because refinancing repays the intermediary;
- equity dynamically complements debt because future refinancing makes current secured debt safer.

Interpretation. Better future equity access increases current debt capacity. This is why public firms or PE-backed firms should, in the paper's logic, find cash-flow-based borrowing easier.

4 Economic logic and what drives the results

4.1 What actually creates cash-flow-based financing?

Three ingredients do the heavy lifting:

- the intermediary can bear at least some risk;
- future equity issuance is feasible often enough;
- distress can be resolved without immediate liquidation.

If any of these fail, financing reverts toward pure collateral lending.

Interpretation. Cash-flow-based lending is not a primitive. It is an equilibrium outcome that depends on institutions and market access.

4.2 Why α is indispensable

The paper explicitly shows that if $\alpha = 0$, the intermediary cannot benefit from the gains of future equity issuance. Then the continuation boundary collapses to the intermediary's autarky value, and financing capacity stops depending on market access.

Interpretation. Sharing *capital-supply risk*—not just ordinary cash-flow risk—is what connects refinancing opportunities to financing capacity.

4.3 Why bankruptcy is more than liquidation

If the firm enters the Chapter 11 region, equity holders may still retain a continuation stake even though creditors control the firm. That stake matters because the firm may earn its way out of bankruptcy or be refinanced later.

Interpretation. Immediate transfer of full ownership to creditors is inefficient. The bankruptcy process is useful precisely because it delays that transfer while preserving optimal risk sharing.

4.4 Why Chapter 11 and Chapter 7 map differently to financing regimes

The paper's mapping is sharp:

- **cash-flow-based financing** is associated with Chapter 11 style continuation and reorganization only;
- **asset-based financing** is associated with both Chapter 11 reorganization and Chapter 7 liquidation.

Interpretation. If cash-flow-based debt exists, it exists because creditors expect the firm to survive distress rather than be liquidated. That is why Chapter-11-like systems are important for this kind of lending.

4.5 What drives the comparative statics in Figure 3

The paper's numerical results imply:

- higher $1/\pi$ (worse equity access) lowers financing capacity and raises precautionary cash;
- higher profitability μ raises financing capacity and generally lowers the need for cash buffers;
- higher volatility σ reduces financing capacity and raises cash holdings;
- higher intermediary risk aversion ρ reduces financing capacity and makes the regime more likely asset-based.

Interpretation. The model says that financing constraints are jointly about firm fundamentals, lender balance-sheet capacity, and outside capital-market conditions.

4.6 Why the pecking order is “overlapping”

The paper does not produce a simple ranking like “cash first, then unsecured debt, then secured debt” in all states. Instead:

- when the firm has cash, it uses cash and unsecured debt together;
- when the firm runs out of cash, it uses unsecured and secured debt together.

Interpretation. Unsecured debt is the universal risk-sharing tool; secured debt is the last-resort funding instrument once internal liquidity is gone.

4.7 What assumptions are doing the most work?

The core assumptions are:

- shareholders are hand-to-mouth and cannot inject funds;
- equity access is random and intermittent;
- cash has a carry cost;
- the intermediary has limited but positive risk-bearing capacity;
- the contract is otherwise complete and there is commitment in the baseline.

Interpretation. These assumptions isolate a very specific friction: liquidity management under constrained equity access. The paper is not fundamentally about managerial moral hazard or covenant incompleteness.

4.8 What the paper can and cannot distinguish

The model is very strong on the *mechanism* linking distress resolution to debt capacity, but it is not an empirical estimation exercise.

- It does not estimate the quantitative importance of the mechanism in data.
- It does not model detailed bargaining among multiple creditor classes.
- It does not emphasize managerial agency or asymmetric information.
- It does, however, generate sharp qualitative predictions about who should get more cash-flow-based debt and when continuation should dominate liquidation.

4.9 Empirical predictions that come out of the model

The paper highlights several predictions:

- firms with better access to equity financing should use more cash-flow-based debt;
- PE-backed firms should obtain higher leverage and be less likely to be liquidated in distress;

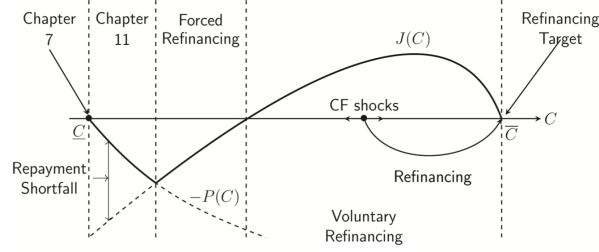


FIGURE 1
Bankruptcy and Control Rights

Notes: Schematic graph of the different regions of the value of refinancing to existing shareholders under optimal (Y, β, α) .

- weaker creditor protection should reduce cash-flow-based debt and shift distress outcomes toward liquidation;
- stronger lender monitoring ability should increase cash-flow-based financing capacity.

5 Figures and propositions

5.1 Figure 1: bankruptcy and control rights

Read. Figure 1 partitions the state space into four regions using $J(C)$:

- **Voluntary Refinancing:** $J(C) > 0$; shareholders willingly refinance and creditors are repaid in full.
- **Forced Refinancing:** $-P(C) < J(C) < 0$; a covenant violation threatens bankruptcy, but refinancing still repays creditors fully and leaves some value for equity.
- **Chapter 11:** $J(C) = -P(C)$; the refinancing constraint binds, creditors force continuation bankruptcy, equity is wiped out if refinancing occurs.
- **Chapter 7:** the credit line is exhausted and the firm liquidates.

Economic message. Bankruptcy is not a side issue. It is the way the model allocates control rights exactly where debt overhang would otherwise block efficient refinancing.

5.2 Proposition 1 and Proposition 2: value function and optimal contract

Read. Proposition 1 reduces the problem to one state variable C and shows that the firm pays dividends only at the upper boundary $\bar{C}$. Proposition 2 characterizes the optimal controls and the lower boundary: either liquidation at $-L$ or continuation at C^S .

Economic message. The optimal contract is a free-boundary problem. One boundary says “pay out,” the other says “liquidate or continue.” The nature of the lower boundary is the essence of asset- versus cash-flow-based finance.

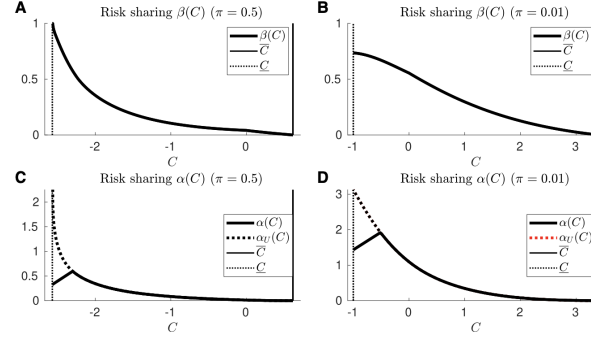


FIGURE 2
Contract Dynamics

Notes: This figure plots $\beta(C)$ and $\alpha(C)$ against C both for $\pi = 0.5$ (Panels A and C) and $\pi = 0.01$ (Panels B and D). The parameters follow Table 1

5.3 Proposition 3 and Corollary 2: implementation via debt

Read. Proposition 3 shows that the cumulative transfer process can be rewritten as a debt balance $T(C)$. Corollary 2 then decomposes that balance into senior-secured debt $Y(C)$ and an unsecured risky credit line $\alpha^U(C)$.

Economic message. Standard-looking securities are not imposed by hand. They are the exact implementation of the optimal long-term contract.

5.4 Figure 2: contract dynamics

Read. Figure 2 plots $\beta(C)$ and $\alpha(C)$ against net liquidity for two refinancing intensities.

- $\beta(C)$ declines with C : risk-sharing through the intermediary is strongest when the firm is most constrained.
- $\alpha(C)$ also declines with C when unconstrained, but near the lower boundary it can increase mechanically because the refinancing constraint binds less as C rises.

Economic message. Both forms of risk sharing are most valuable in distress, but capital-supply risk sharing is precisely where limited liability bites.

5.5 Figure 3: determinants of financing capacity and cash policy

Read. Figure 3 studies how the upper and lower boundaries move with refinancing access, profitability, volatility, and intermediary risk aversion.

Economic message. The figure turns the theory into concrete comparative statics: cash-flow-based debt is easier to sustain when firms are better and lenders are stronger.

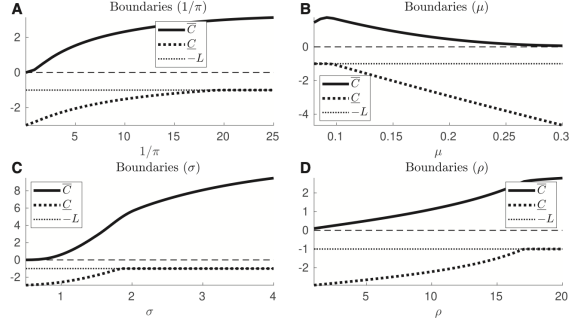


FIGURE 3
Boundaries

Notes: Comparative statics of the boundaries $\bar{C}$, $\bar{C}$ with respect to the expected time until refinancing $1/\pi$ (top left panel), profitability μ (top right panel), cash flow volatility σ (bottom left panel), intermediary risk aversion ρ (bottom right panel). The parameters follow Table 1

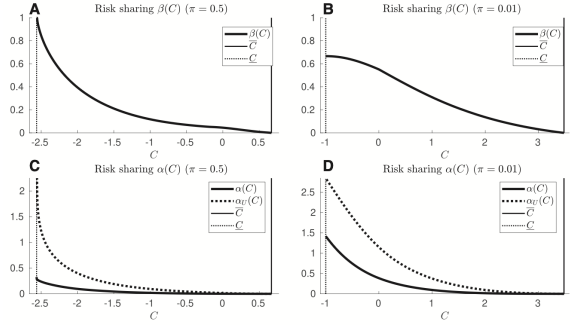


FIGURE 5

Contract Dynamics under limited creditor protection

Notes: This figure plots $\beta(C)$ and $\alpha(C)$ against C both for $\pi = 0.5$ (Panels A and C) and $\pi = 0.01$ (Panels B and D). The parameters follow Table 1 and $\nu = 0$ in refinancing constraint (33)

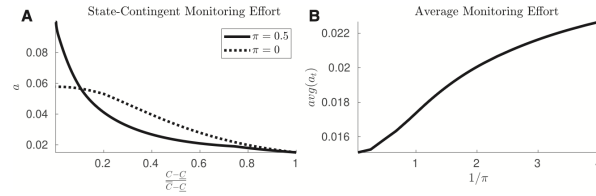


FIGURE 6

Intermediary Incentives

Notes: The left panel of this figure plots the intermediary's effort $\alpha = \beta/\kappa$ both for $\pi = 0.5$ (solid black line) and for $\pi = 0$ (dotted red line) in the transformed state space against $(\bar{C} - C)/(\bar{C} - \underline{C})$. The right panel plots average effort $\text{avg}(\alpha_t)$ against $1/\pi$. The parameters follow Table 1, and we use $\kappa = 10$

5.6 Figure 5 and Proposition 4: weak creditor protection

Read. The extension replaces the baseline refinancing constraint with

$$\alpha(C) \leq [P(\bar{C}) - \bar{C}] - [(1 - \nu)P(C) - C] \iff J(C) \geq -\nu P(C), \quad (30)$$

where $\nu \in [0, 1]$ captures shareholders' commitment to refinance and, equivalently, creditor protection.

Proposition 4 shows that lower ν reduces both the payout boundary and the continuation boundary.

Economic message. Weak creditor rights reduce cash-flow-based financing capacity and move the economy toward liquidation-based outcomes. The paper's mechanism therefore depends in an important way on creditor protection.

5.7 Figure 6: intermediary incentives and monitoring

Read. In the monitoring extension, cash flows become

$$dX_t = (\mu + a_t)dt + \sigma dZ_t, \quad (31)$$

with intermediary effort cost $\kappa a_t^2/2$. The incentive condition is

$$a_t = \frac{\beta_t}{\kappa}. \quad (32)$$

Monitoring effort rises when the intermediary has more skin in the game.

Economic message. Cash-flow-based financing is not only about giving the lender claims on future cash flow; it also gives the lender incentive to monitor, especially in distress.

6 Short summary

- The paper studies a firm that manages liquidity risk with costly cash and costly intermediary finance while outside equity arrives only intermittently.
- The sufficient state variable is net liquidity, $C = M - Y$.
- The optimal contract solves an HJB with an upper payout boundary and a lower distress boundary.
- If the lower boundary is liquidation, financing is asset-based; if the intermediary optimally keeps the firm alive, financing is cash-flow-based.
- The contract is implemented by senior-secured debt plus an unsecured risky credit line.
- Bankruptcy is an optimal risk-sharing institution: Chapter 11 reallocates risk without immediate liquidation, while Chapter 7 liquidates once financing capacity is exhausted.
- Unsecured debt protects secured debt, so the two are complements.
- Better access to equity financing increases current debt capacity, so debt and equity are dynamic complements.
- Weak creditor protection or weak intermediary risk-bearing capacity pushes the system back toward asset-based finance and liquidation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes a dynamic microfoundation for why some firms borrow only against collateral while others borrow against expected cash flows. The boundary between those two cases is not arbitrary: it is endogenously determined by the interaction of firm cash flows, lender risk-bearing capacity, future equity access, and the way distress is resolved.

7.2 Economic intuition

The most important intuition is circular in a productive way. Creditors are willing to lend against future cash flows only if they expect the firm to survive distress long enough for those cash flows to arrive or for refinancing to occur. But the firm survives distress only because creditors are willing to lend against those same future cash flows. Chapter-11-style continuation closes that loop.

7.3 Takeaway

This paper is ultimately about the idea that *financing capacity is created by institutions*. A liquidation-only world produces collateral lending. A world with creditor control, partial default on junior claims, and continuation in distress can produce true cash-flow-based financing.